

BONAM MINGOLE

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Ph.D. Candidate ◇ Language Models Finetuning ◇ University Park, PA, USA

EDUCATION

Ph.D. in Informatics Pennsylvania State University <i>Research focus:</i> Language Models Finetuning <i>Advisor:</i> Prof. Amulya Yadav	State College, USA <i>Aug. 2024 - Present</i>
M.S. in Software Engineering University of Quebec in Montreal <i>MS Thesis:</i> Analysis and co-creation of an AI assistant in the Financial Industry	Montreal, Canada <i>Sept. 2017 - Jan. 2020</i>
B.B.A in Business Technologies University of Quebec in Montreal <i>International exchange student:</i> EMS Strasbourg, France.	Montreal, Canada <i>Sept. 2013 - May 2017</i>

TECHNICAL SKILLS

Programming Languages	Python, SQL, JavaScript, HTML, Bash
Framework & Tools	LLMs, RAG, Sci-Kit Learn, Pytorch, React, FastAPI, LangChain, Git

WORK EXPERIENCE

AI Research Assistant RAISE Lab @ Penn State University <i>Finetuning and Evaluating Language models for safer daily use</i>	Stage College, USA <i>Aug. 2024 - Now</i>
• Built and deployed LLM Finetuning for Adaptive Academic Tutoring. See ALAASKA Repository • Led research studies to assess LLMs limitations including biases, misinformation, and impact on higher education. • implemented Retrieval-Augmented Generation (RAG) pipelines to improve LLM accuracy from 80 to 90% for medical diagnostic on crowdsourced patient cases. See DAT RAG Repository. • Audited LLM fairness by comparing model accuracy (F1 score) between geographical regions.	
Humanitarian AI Research Intern United Nations OCHA <i>AI for Humanitarian Analysis and Crisis Response</i>	The Hague, Netherlands <i>Jun - Aug 2025</i>
• Conducted research on the use of AI tools to support humanitarian analysis for crisis response planning. • Built fine-tuned LLMs leveraging RAG and prompt engineering for accurate organizational information retrieval. • Supported the OCHA-AI community by documenting best practices and AI toolkits for field staff. • Drafted internal reports and recommendations on responsible AI adoption for humanitarian action. • Collaborated across UN branches to align AI initiatives with humanitarian principles and data protection standards.	
AI Research Assistant Mila Quebec AI Institute <i>Responsible AI operationalization through Metrics</i>	Montreal, Canada <i>Aug. 2023 - Aug. 2024</i>
• Conducted a <i>Scoping Review</i> on how AI researchers operationalize ethical principles through measurement. • Extracted data from 257 papers (from 2011 - 2023) measuring AI systems adherence to Responsible AI principles. • Created the Responsible AI Measures Dataset, consolidating 11,865 data points across 791 evaluation measures, covering 11 ethical principles including Fairness, Transparency, Privacy, Trust, Sustainability, and more. • Developed a taxonomy of metrics and evaluation methods for Responsible AI principles.	
AI Research Fellow AI4Good Lab <i>Applied NLP Research and AI Policy</i>	Montreal, Canada <i>May 2023 - Aug. 2023</i>
• Developed AITHICAL, a prototype ML platform that aggregates AI ethics guidelines and regulations and suggests regulatory guidance to AI practitioners based on their projects descriptions. See AITHICAL GitHub Repo • Trained and tested a zero-shot NLP model to classify AI projects into 6 categories targeted by Canadian AI regulations. • Initiated a mentorship program to create a more inclusive and diverse Tech culture in Canada.	

AI Research Partnerships Coordinator [Part-time] | Mitacs Canada
AI Research Governance

Montreal, Canada
Jan. 2022 - Aug. 2024

- Evaluated 10,000+ AI research grant proposals, contributing to the allocation of \$50M+ in research funding.
- Designed a structured evaluation framework that reduced review time by 20% while maintaining quality.
- Facilitated partnerships between academia, industry, and nonprofits, enabling 130+ human-centered AI initiatives.
- Analyzed proposals trends to identify emerging AI research areas, guiding future funding strategies.

AI Solutions Analyst | Intact Insurance - Data Lab
Applied Machine Learning and AI deployment

Montreal, Canada
May 2021 - Apr. 2023

- Coordinated the development and deployment of XGBoost regression models to predict insurance premiums for 500K+ commercial and personal policies, achieving 98% model accuracy.
- Deployed NLP solutions to summarize insurance claims, enhancing efficiency for 5k+ employees.
- Conducted A/B testing, model validation, and user acceptance testing ensuring 95% user satisfaction rate.

AI Solutions Analyst | National Bank of Canada
Participatory design for NLP Solutions

Montreal, Canada
Jun. 2018 - Apr. 2021

- Used participatory design to implement an Azure-based chatbot for HR inquiries, serving 7K+ employees across Canada.
- Gathered and managed requirements for a smart search engine, enhancing information retrieval accuracy by 15%.
- Translated AI research findings into actionable insights aligned with organizational digitalization goals.

PUBLICATIONS

7. Mingole, B. et al. (2026). *The Cheat-a-thon: How can large language models impact higher education?*. Under review - ACM 2026.  Media Coverage: [Penn State Cheat-a-thon](#)
6. **Mingole, B.**, Majumdar A., Choudhury F. et al. (2025). *Dr. GPT Will See You Now, but Should It? Exploring the Benefits and Harms of LLMs for Health Advice using Crowdsourced Clinical Cases*. Under review - [FAccT 2026](#).
5. Wang, X., **Mingole, B.** et al. (2024). *Have LLMs Reopened the Pandora's Box of AI Fake News?*. [NAACL 2025](#).
4. Guo, H., **Mingole, B.** et al. (2025). *Exposing AI Bias by Crowdsourcing: Democratizing Critique of Large Language Models*. [AIES 2025](#).
3. Rismani, S., Davis, L., **Mingole, B.** et al. (2025) *Responsible AI measures dataset for ethics evaluation of AI systems*. [Nature Scientific Data 2025](#).
2. **Mingole, B.** et al. (2023). *AITHICAL: an AI Ethics guidelines aggregator Prototype*. [AI4Good Lab 2023](#).
1. **Mingole, B.** (2020). Analysis and Co-creation of an AI assistant in the financial industry [Master's thesis, University of Quebec in Montreal]. [Archipel UQAM](#).

MACHINE LEARNING (ML) AND AI PROJECTS

DigitalAIze | *Collection of ML applications to facilitate Digital Transformation*

[DigitalAIze Web App](#)

Set of Online Machine learning applications aimed at enabling Digital Transformation for small businesses in communities with limited access to information technology.

Oct 2023

MRAW | *Prediction of Life Expectancy based on Access to Clean Drinking Water*

[MRAW Web App](#)

Online ML solution using XGboost regression to predict life expectancy in a country based on the percentage of people accessing clean drinking water.

Jul 2022

TEACHING EXPERIENCE

Teaching Assistant

- DS 442 (PSU) | Artificial Intelligence Foundations
- INF 2050 (UQAM) | Software Development Tools and Practices
- INF 1120 (UQAM) | Introduction to Programming

Aug. 2025 - Dec. 2025

Jan. 2019 - May 2019

Sep. 2018 - Dec. 2018